

PROJECT DESIGN

3. PROJECT DESIGN PHASE

3.1 SYSTEM DESIGN

The project is designed to implement Script-Controlled Access Control Lists (ACLs) in ServiceNow. The system controls access to records based on the user's assigned role and the value of the Branch field.

3.2 SYSTEM COMPONENTS

The project consists of the following main components:

ServiceNow Platform

EEE User

Custom Roles: bb1, bb2, bb3, bb4

Institution Details Table

Student Records

Read ACL

Create ACL

Write ACL

Delete ACL

The project instructions define the Institution Details table and the four custom roles as part of the implementation.

3.3 ACCESS CONTROL DESIGN

The access control is designed using different ACL operations:

ACL Operation	Required Role
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Read	bb1
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Create	bb2
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Write	bb3
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Delete	bb4
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The Read ACL also uses the Branch is EEE data condition.

3.4 ACCESS CONTROL FLOW

The basic access flow is:

User → Role Verification → ACL Check → Branch/Access Condition → Allow or Deny Access

The system checks the user's role and the configured ACL conditions before providing access to the records.

3.5 TABLE DESIGN

The main table used in the project is:

Table Label: Institution Details

Table Name: u_institution_details

The table contains student-related fields including Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, and Description.

3.6 SECURITY DESIGN

The security design ensures that:

Users are given access according to their assigned roles.

EEE branch records are protected through the Read ACL condition.

Create, Write, and Delete operations are controlled using separate ACLs.

Administrators retain full access to the records.

3.7 EXPECTED DESIGN OUTCOME

The designed system provides role-based and field-based record-level security in ServiceNow. It controls viewing, creation, modification, and deletion of records through ACLs.